```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
df = pd.read_csv("netflix_titles.csv")
df.head()
```

	show_id	type	title	director	cast	country	date_added	release_year	rating	duration	listed_in	description
0	s1	Movie	Dick Johnson Is Dead	Kirsten Johnson	NaN	United States	September 25, 2021	2020	PG-13	90 min	Documentaries	As her father nears the end of his life, filmm...
1	s2	TV Show	Blood & Water	NaN	Ama Qamata, Khosi Ngema, Gail Mabalane, Thaban... Sami	South Africa	September 24, 2021	2021	TV-MA	2 Seasons	International TV Shows, TV Dramas, TV Mysteries	After crossing paths at a party, a Cape Town t...

Next steps: Generate code with df New interactive sheet

DATA PREPROCESSING

```
df = df.dropna(subset=['listed_in', 'rating'])
df = df[['type', 'listed_in', 'release_year', 'rating']]

df['Genre'] = df['listed_in'].apply(lambda x: x.split(',')[0])
```

SIMULATION

```
import numpy as np

np.random.seed(42)

df['Age'] = np.random.randint(16, 30, size=len(df))
df['Watch_Time'] = np.random.randint(1, 10, size=len(df))
```

ADVANCED SIMULATION

```
genre_weights = {
    'Drama': 0.3,
    'Comedy': 0.25,
    'Action': 0.2,
    'Thriller': 0.15,
    'Romance': 0.1
}

df['Preference_Score'] = df['Genre'].map(genre_weights).fillna(0.05)

df['Preference_Score'] += np.random.rand(len(df)) * 0.1
```

PROBABILITY

```
prob = df['Genre'].value_counts(normalize=True)
print(prob)
```

```
Genre
Dramas                        0.181643
Comedies                      0.137453
Action & Adventure            0.097580
Documentaries                 0.094172
International TV Shows        0.087925
Children & Family Movies      0.068727
Crime TV Shows                0.045325
```

```
Kids' TV                        0.043962
Stand-Up Comedy                 0.037942
Horror Movies                   0.031239
British TV Shows                0.028740
Docuseries                      0.025105
Anime Series                    0.019880
International Movies            0.014540
TV Comedies                     0.013632
Reality TV                      0.013632
Classic Movies                  0.009088
TV Dramas                       0.007611
Thrillers                       0.007384
Movies                          0.006361
TV Action & Adventure           0.004544
Stand-Up Comedy & Talk Shows    0.003862
Romantic TV Shows               0.003635
Classic & Cult TV               0.002499
Anime Features                  0.002386
Independent Movies              0.002272
Music & Musicals                0.002045
TV Shows                        0.001818
Sci-Fi & Fantasy                0.001477
Cult Movies                     0.001363
TV Horror                       0.001250
Romantic Movies                 0.000341
Spanish-Language TV Shows       0.000227
LGBTQ Movies                    0.000114
TV Sci-Fi & Fantasy             0.000114
Sports Movies                   0.000114
Name: proportion, dtype: float64
```

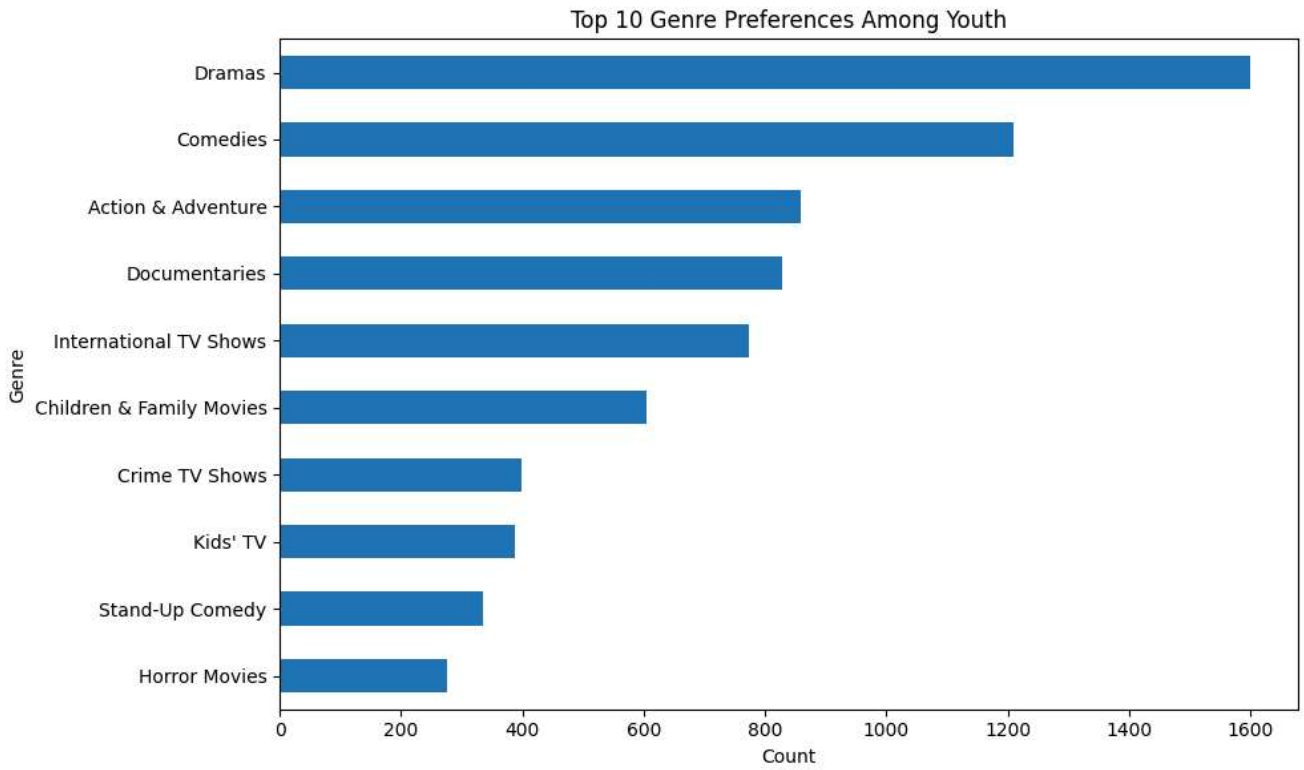
STATISTICS

```
print("Mean Age:", df['Age'].mean())
print("Median Watch Time:", df['Watch_Time'].median())
print("Variance:", df['Watch_Time'].var())
```

```
Mean Age: 22.498693627172553
Median Watch Time: 5.0
Variance: 6.64681957719132
```

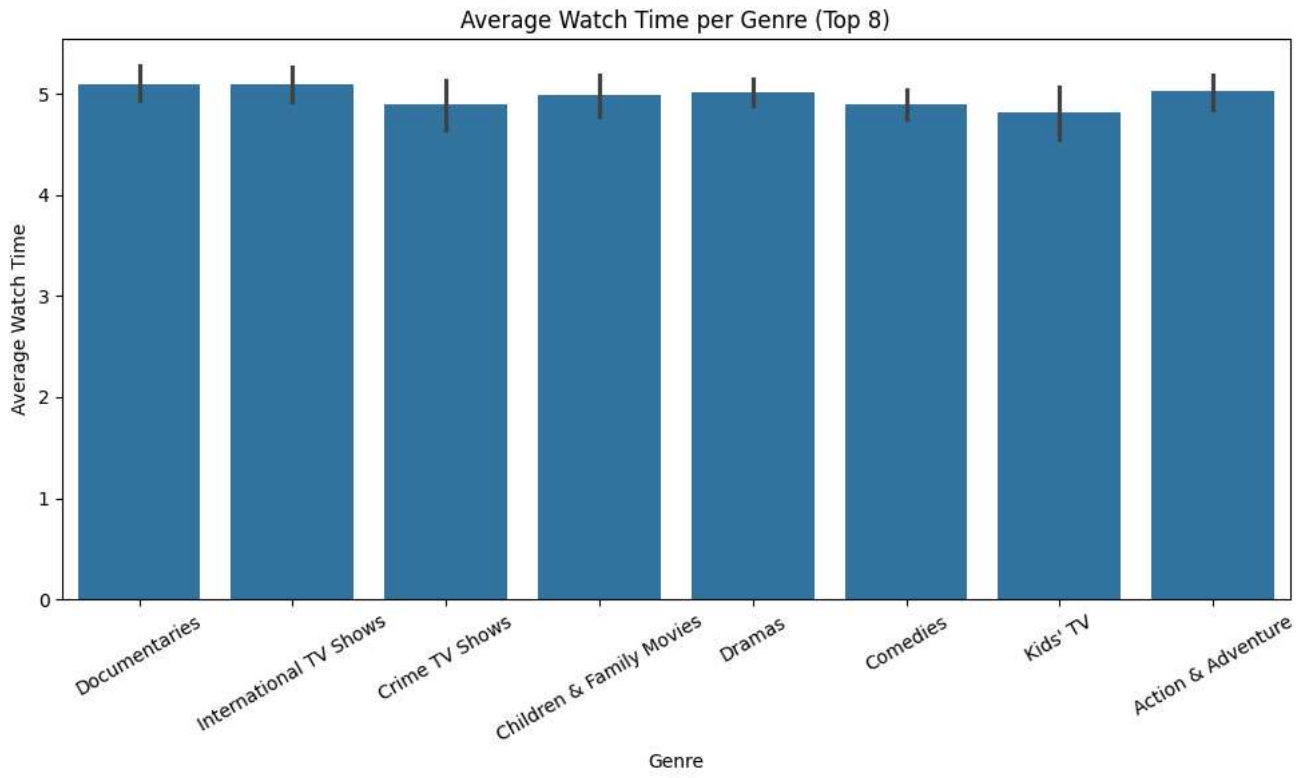
PREFERNCES TOP 10 GENRES ONLY

```
top_genres = df['Genre'].value_counts().head(10)

plt.figure(figsize=(10,6))
top_genres.sort_values().plot(kind='barh')

plt.title("Top 10 Genre Preferences Among Youth")
plt.xlabel("Count")
plt.ylabel("Genre")
plt.tight_layout()

plt.show()
```



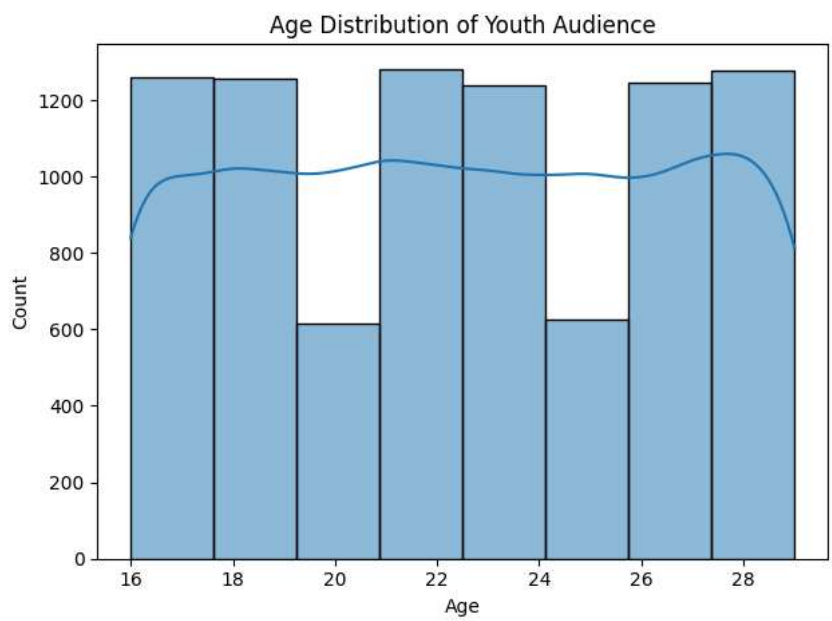
Watch Time vs Genre (TOP 8)

```
top8 = df['Genre'].value_counts().head(8).index
df_top = df[df['Genre'].isin(top8)]

plt.figure(figsize=(10,6))
sns.barplot(x='Genre', y='Watch_Time', data=df_top)

plt.title("Average Watch Time per Genre (Top 8)")
plt.xlabel("Genre")
plt.ylabel("Average Watch Time")
plt.xticks(rotation=30)
plt.tight_layout()

plt.show()
```



AGE DISTRIBUTION

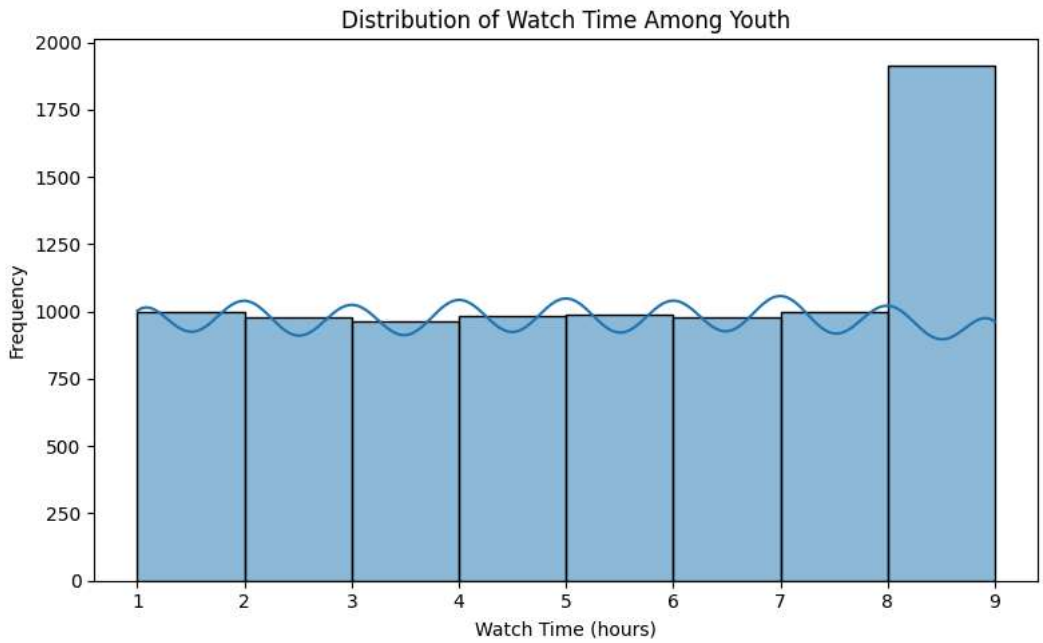
```
sns.histplot(df['Age'], bins=8, kde=True)

plt.title("Age Distribution of Youth Audience")
plt.xlabel("Age")
plt.ylabel("Count")
plt.tight_layout()
plt.show()
```



Watch Time Distribution (EDA)

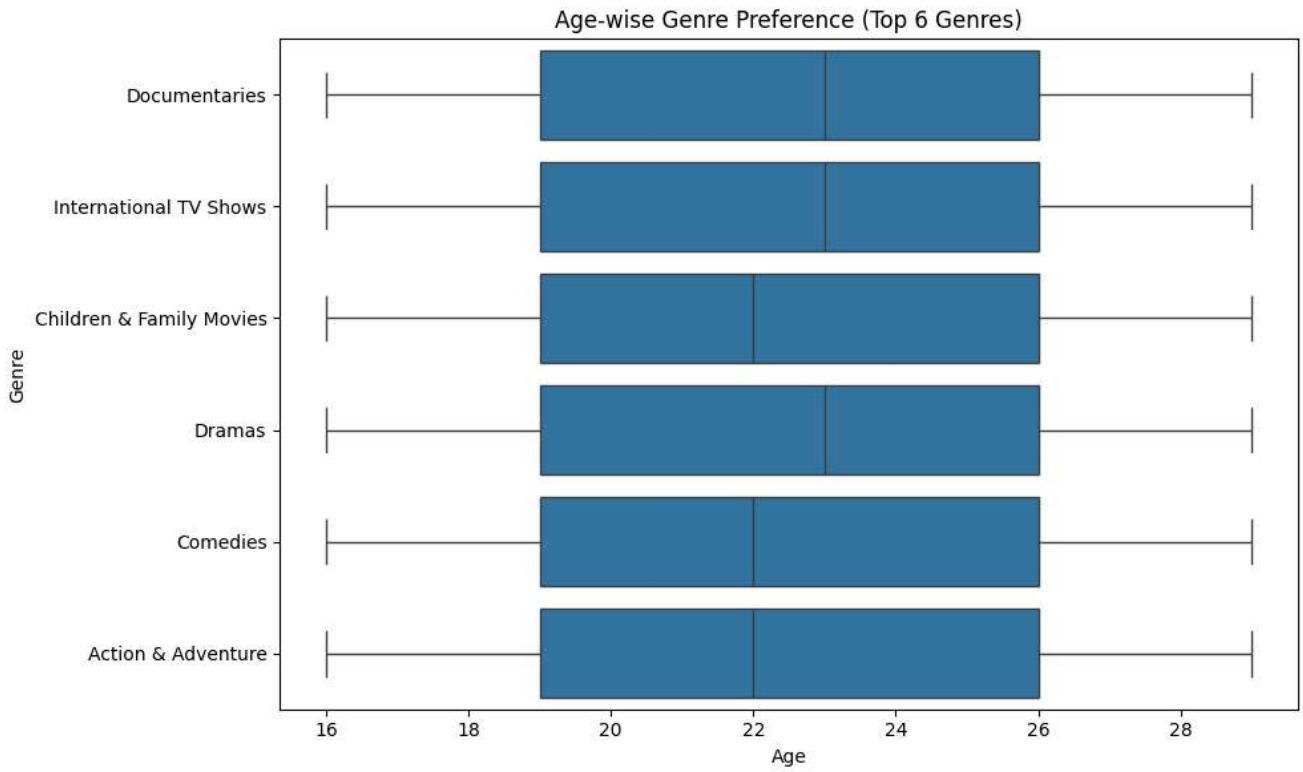
```
plt.figure(figsize=(8,5))

sns.histplot(df['Watch_Time'], bins=8, kde=True)
```

```
plt.title("Distribution of Watch Time Among Youth")
plt.xlabel("Watch Time (hours)")
plt.ylabel("Frequency")
plt.tight_layout()
plt.show()
```



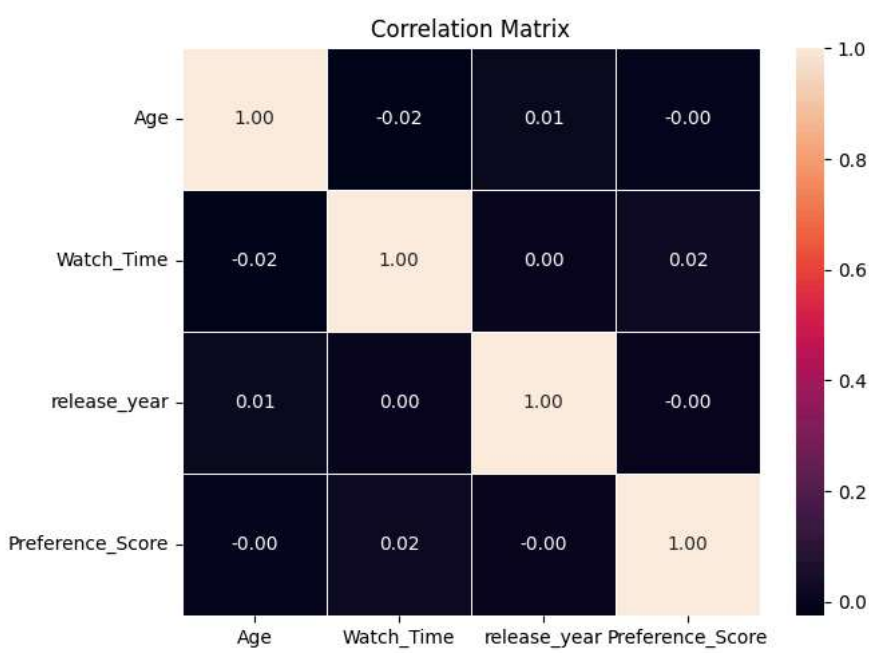
Age vs Genre (Box Plot)

```
top6 = df['Genre'].value_counts().head(6).index
df_top = df[df['Genre'].isin(top6)]

plt.figure(figsize=(10,6))
sns.boxplot(y='Genre', x='Age', data=df_top)

plt.title("Age-wise Genre Preference (Top 6 Genres)")
plt.xlabel("Age")
plt.ylabel("Genre")
plt.tight_layout()
plt.show()
```



CORRELATION HEATMAP

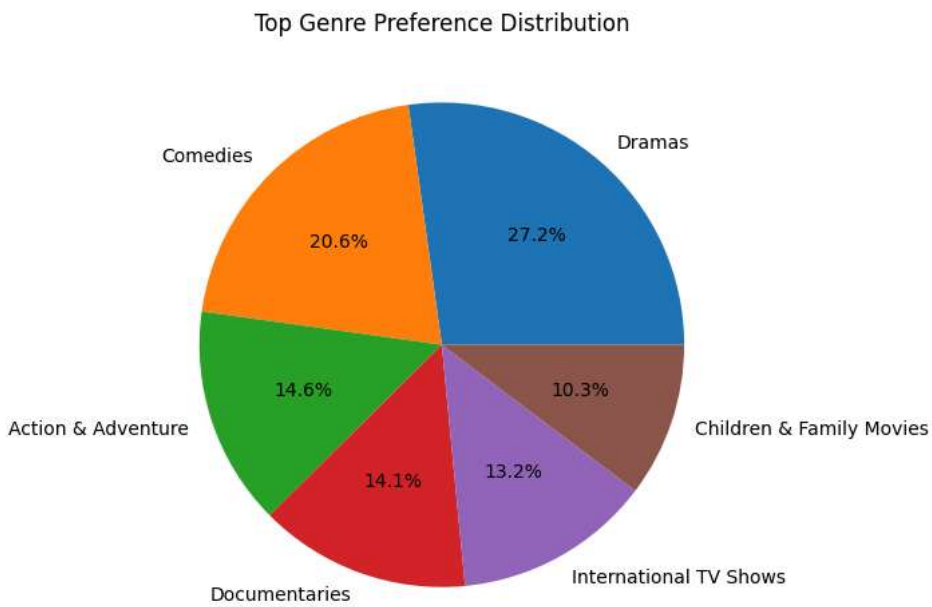
```
corr = df[['Age','Watch_Time','release_year','Preference_Score']].corr()

plt.figure(figsize=(7,5))
sns.heatmap(corr, annot=True, fmt=".2f", linewidths=0.5)

plt.title("Correlation Matrix")
plt.tight_layout()
plt.show()
```



PROBABILITY PIE CHART

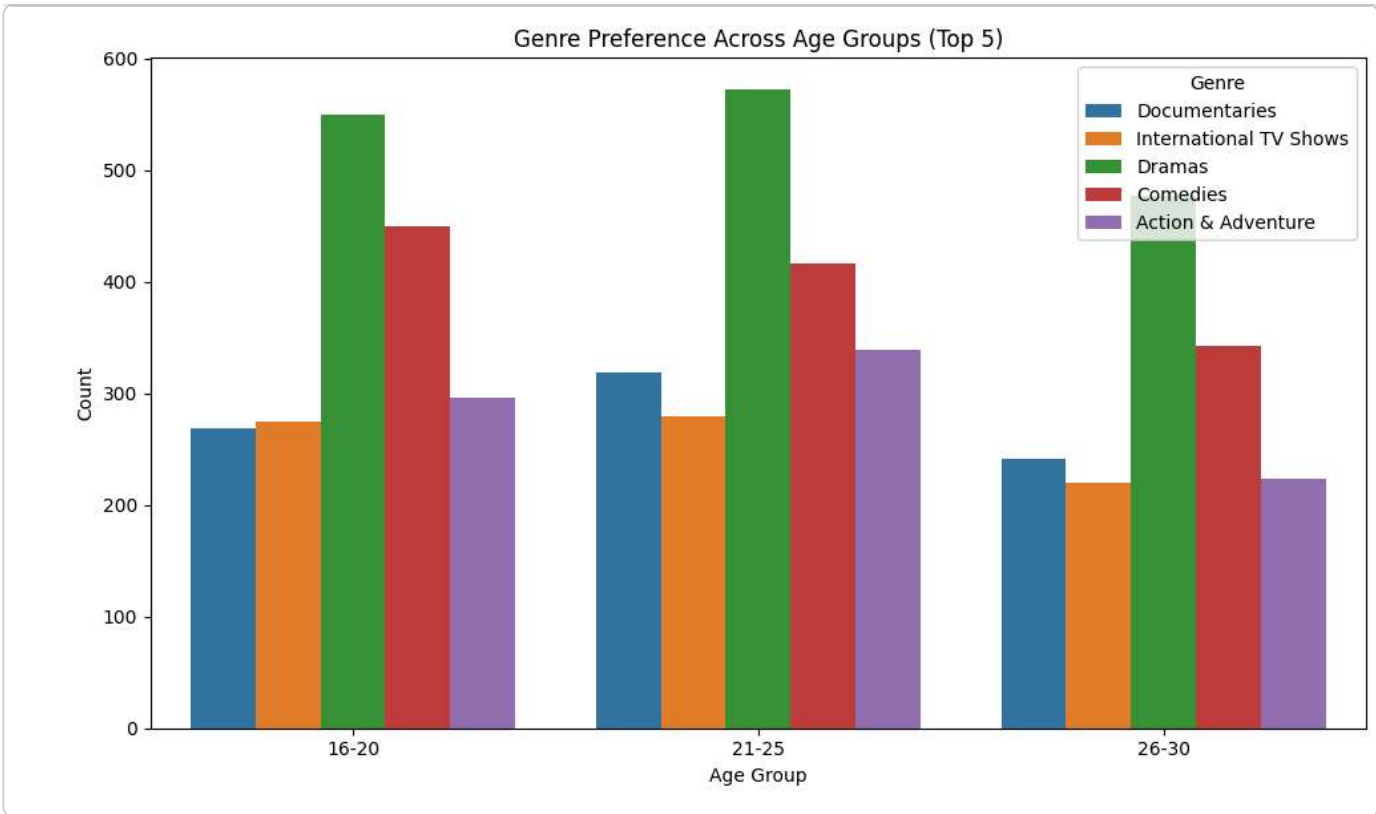
```
genre_prob = df['Genre'].value_counts(normalize=True).head(6)
```

```
plt.figure(figsize=(7,7))
genre_prob.plot(kind='pie', autopct='%1.1f%%')

plt.title("Top Genre Preference Distribution")
plt.ylabel("")
plt.tight_layout()
plt.show()
```



AGE GROUP vs GENRE

```
df['Age_Group'] = pd.cut(df['Age'], bins=[15,20,25,30], labels=['16-20','21-25','26-30'])

top5 = df['Genre'].value_counts().head(5).index
df_top = df[df['Genre'].isin(top5)]

plt.figure(figsize=(10,6))
sns.countplot(x='Age_Group', hue='Genre', data=df_top)

plt.title("Genre Preference Across Age Groups (Top 5)")
plt.xlabel("Age Group")
plt.ylabel("Count")
plt.legend(title="Genre")
plt.tight_layout()
plt.show()
```



Predictions

```
y_pred = model.predict(X_test)
```

Prediction genre preference

```
from sklearn.preprocessing import LabelEncoder

le = LabelEncoder()
df['Genre_encoded'] = le.fit_transform(df['Genre'])

X = df[['Age', 'Watch_Time', 'release_year']]
y = df['Genre_encoded']
```

Trained model

```
from sklearn.ensemble import RandomForestClassifier

model = RandomForestClassifier(random_state=42)
model.fit(X_train, y_train)
```

```
RandomForestClassifier
RandomForestClassifier(random_state=42)
```

Accuracy

```
from sklearn.metrics import accuracy_score

accuracy = accuracy_score(y_test, y_pred)
print("Accuracy:", accuracy)
```

```
Accuracy: 0.4934696195343555
```

High vs Low Preference Target

```
df['Preference_Label'] = (df['Preference_Score'] > df['Preference_Score'].mean()).astype(int)
```

```
df[['Preference_Score', 'Preference_Label']].head()
```

	Preference_Score	Preference_Label
0	0.087426	0
1	0.138411	1
2	0.116766	1
3	0.070541	0
4	0.081990	0

Defining Features & Target

```
X = df[['Age', 'Watch_Time', 'release_year']]
y = df['Preference_Label']
```

Train-Test Split

```
from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
```